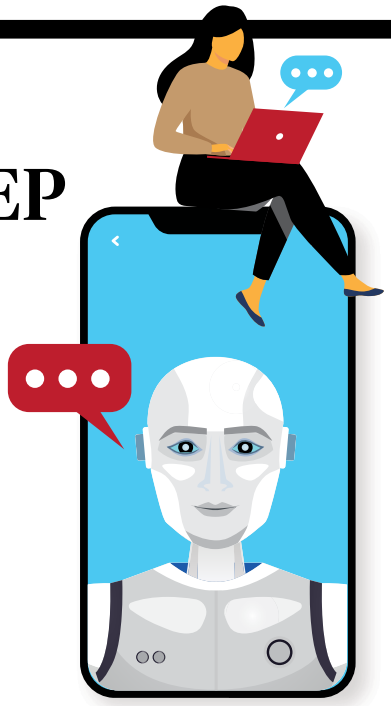
IT AND DEEP TECH

TAKE THE DEEP TECH LEAP



India stands at a critical juncture in its technological evolution. As the world races ahead in artificial intelligence (AI), quantum computing and semiconductors, the government must take decisive steps to propel the country into the next era of digital leadership. Developing a strong AI ecosystem is a priority. AI's potential spans industries, from healthcare and environmental sustainability to agriculture and e-governance. Central to this effort is the deployment of Graphics Processing Units (GPUs), the powerhouses of AI computation. However, India remains in the early stages of AI infrastructure development. To bridge this gap, a public-private partnership model must be aggressively pursued to ensure that the country has access to cutting-edge processing power.

India is currently working on multiple foundational AI models set for deployment within the next year. Yet, a major bottleneck is the availability of digital-first datasets tailored to Indian use cases. The solution lies in large-scale community-sourced datasets. However, experts warn that for such projects to be truly open source, all four essential components—algorithms, datasets, model weights and source code—must be accessible for public review and use. Additionally, Indian AI models must prioritise algorithmic efficiency to reduce development costs and delays to compete with global counterparts. To fast-track this goal, India is in discussions with three global technology giants to design and develop indigenous GPUs. These may not rival the world's most advanced processors but could serve as cost-effective alternatives that encourage innovation in AI applications.

Even as govt explores a Deep Tech Fund of Funds to catalyse the next-gen startups, a concrete roadmap is awaited

Several Indian states, particularly in the south, have taken the lead in AI adoption, even ahead of the national India AI Mission launched last year. However, beyond policy updates, India must significantly expand AI skilling initiatives and reinvigorate investments in AI startups. In 2024, Indian AI startups raised only $166 million (Rs 1,445 crore at current exchange rate)—a sharp decline from $518 million (Rs 4,510 crore) in 2022.

GIVE STARTUPS A LEG-UP

Recognising the importance of deep tech, finance minister Nirmala Sitharaman in her budget speech proposed exploring a Deep Tech Fund of Funds "to catalyse the next generation startups". However, industry body Nasscom had recommended a more concrete Rs 10,000 crore fund to fuel early-stage startups. The fund, as per Nasscom, should target two critical areas. The first would support high-growth sectors like EV batteries, space tech, biotech and select AI firms—helping Indian startups compete with US and Chinese counterparts. The second would focus on strategic technologies such as defence, agricultural biotech and quantum computing—areas that may not have immediate commercial potential but are crucial for long-term technological sovereignty. While the government has initiated promising measures, a far bolder approach is needed to build a robust IT and deep tech ecosystem. ■

LABOUR

STOP LABOURING OVER REFORM



India has a burgeoning workforce. At last count, it stood at 565 million, with 55 per cent engaged in manufacturing, construction and service sectors. The formal sector has seen significant growth, with net Employees' Provident Fund Organisation (EPFO) subscriptions more than doubling from 6.1 million in FY19 to 13.1 million in FY24, states the Economic Survey. According to the NITI Aayog, even the number of gig and platform workers—who undertake temporary, flexible jobs, often facilitated by digital platforms—is expected to rise to 23.5 million by FY30. In fact, the labour force participation rate (LFPR), which is the percentage of working-age people engaged in or seeking employment, rose from 49.8 per cent in FY18 to 60.1 per cent in FY24, as per the latest Periodic Labour Force Survey.